



Trenched microwave resonator integrated with porous PDMS for detection and classification of VOCs with enhanced performance

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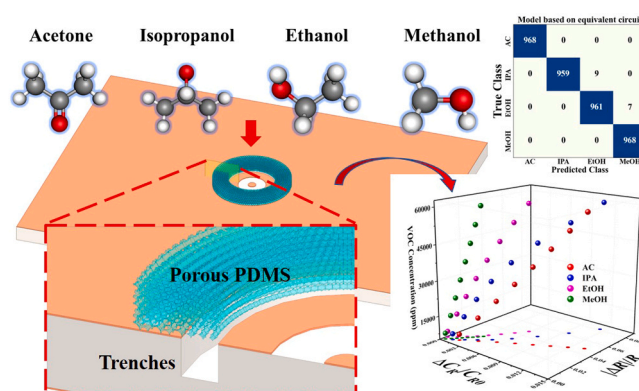
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HIGHLIGHTS

- The improving response speed owing to the porous structure of the absorption layer.
- The improving sensitivity by structural optimization involving the reduction of resonator's substrate capacitance.
- Linear detection of VOCs within a wide concentration range from 10^2 to 10^4 ppm.
- Multiple VOCs classification with a high accuracy of 99.59%.

GRAPHICAL ABSTRACT



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ABSTRACT

Microwave resonators combined with polymer absorption layers are widely used in volatile organic compound (VOC) detection based on their variable resonant frequencies. However, the response time is limited due to the polymer's slow volumetric absorption of VOC molecules. By constructing a porous structure in Polydimethylsiloxane (PDMS), resulting in reduced the response time to as short as 71.1%. To mitigate the sensitivity decline caused by the porous PDMS, a trenched-substrate complementary split-ring resonator (CSRR) is proposed for enhancing the interaction between the electromagnetic fields (EMFs) and the porous PDMS with VOCs. The removal of the substrate beneath CSRR's sensing region enhances the effective EMF, increasing frequency and amplitude sensitivities up to 175.5% and 137.8%, respectively. Responses to four common VOCs by the sensor show a maximum sensitivity of 217 Hz/ppm and a minimum limit of detection of 295 ppm. Additionally, resonant parameters and extracted lumped parameters are utilized to establish two decision-tree-based VOC classification models, achieving high accuracies of 98.71% and 99.59%, respectively. And the latter one fully utilizing responses throughout the swept band, proves superior in identifying similar substances. This sensor technology helps promote the sensitive detection and accurate classification of diverse VOCs.

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1. Introduction

Volatile organic compounds (VOCs) are major contaminants posing severe threats to human health and the environment due to their toxicity, carcinogenicity, and teratogenicity [1–5]. As one primary method, microwave resonator-based VOC sensors utilize interaction between locally concentrated electromagnetic field (EMF) and VOC vapor molecules concentrated by analyte-sensitive material integrated into the sensing region. Among various microwave resonators, split-ring resonators (SRRs) and complementary SRRs (CSRRs) with planar structures have attracted increasing attention owing to their low cost [6], compact size [7], easy implementation [8], and inherent contactless and remote sensing capabilities [9]. The dielectric properties of VOC vapors can be characterized by processing resonance signals of such planar resonators, including resonant frequencies and scattering parameters, and further used for qualitative and quantitative detection of VOCs. To enhance the dielectric response of resonators, various analyte-sensitive materials are widely used, such as graphene, carbon nanotubes, and polymers, among which polymer-based sensitive materials are favorable choice in VOC detection with notable advantages such as ease of fabrication [10], water resistance [11], high stability [12], etc. Nevertheless, microwave sensors incorporated with polymers suffer from the longer response time than other 2D materials due to volumetric absorption rather than surface adsorption [13].

The response time of a microwave VOC sensor mainly depends on the rate of interaction between the analyte molecules and the analyte-sensitive material. Since temperature, related to gas-sensing kinetics, is one of the factors influencing gas response time [14], a gas sensor based on a composite polymer film was affixed to a heater, realizing shorter response time at an operating temperature of 40°C [15]. However, high working temperatures imply shortened lifetime and increased power consumption. By embedding Fe_3O_4 nanoparticles, the diffusion coefficient of VOC molecules within a polyethyleneimine polymer was increased, achieving a maximum 40% reduction in response time compared to the pure polymer film at room temperature [16]. Nevertheless, to ensure the favorable compatibility and dispersibility of nanoparticles within the polymer matrix, meticulous selection of appropriate process parameters is imperative [17]. In contrast, optimizing polymer morphology is a simpler and more versatile approach to promote the response rate. For instance, by using the sacrificial template, the interconnected porous structure was fabricated inside the Polydimethylsiloxane (PDMS), resulting in a 66% reduction in the swelling cycle time for acetone [18]. However, since polymers typically work by volumetric absorption, this approach would result in fewer VOC molecules being diffused into polymer-based sensitive materials, leading to a reduced sensor response magnitude. Therefore, it is on-demand to develop a resonator with higher sensitivity to incorporate porous polymers for rapid VOC detection.

Based on the interaction mechanism between EMF and lossy materials, numerous approaches were proposed based on resonator structure design to enhance dielectric measurements' sensitivity by increasing the fringing electric field strength in the sensing region. For instance, by integrating multiple SRRs concentrically coupled with the micro-strip line, a strong electric field up to 10^5 V/m was achieved at the resonance due to an increase of distributed capacitances, resulting in a 1.3 times improvement in sensitivity to surrounding materials [19]. In addition, inter-digital capacitor (IDC) topology was used to confine the electric field along the interdigital structure [20] and increase the gap capacitance by having a larger effective plate area, contributing to an augmented electric field concentration in the gap of the resonator [21]. These advances in obtaining higher sensitivity measurements aim to increase the total electric field intensity. However, it is noteworthy that reducing the capacitance of non-sensing elements in the resonator structure is equivalent to enhancing the electric field in the sensing region. Utilizing a thin metal trace connecting the microstrip transmission line to the capacitive patch, parasitic coupling capacitance is

successfully removed, demonstrating a superior sensitivity for a wide permittivity range [22]. Similarly, a pair of triangular ground planes were incorporated into the sensor to eliminate leaky EMF in non-sensing regions, thereby enabling highly sensitive detection of the target substance [23]. Moreover, since part of the fringing electric field is concentrated in the substrate layer, the sensitivity can be enhanced by reducing the polarization portion of the substrate through substrate thinning to strengthen the resonating electric field in the materials under test [24]. Similarly, substrate materials with lower dielectric constants can also be used to increase the sensitivity [25]. Aiming to further mitigate the influence of the substrate, the analyte channel was directly integrated into the inner substrate by implementing the stepped structure [26]. Similar approaches have also been applied to the evaluation of blood sugar [27] and characterization of mixed liquids [28]. Nevertheless, the influence of the substrate on the sensitivity has not been decreased to a minimum due to the existence of an equivalent through-substrate capacitor coupled in the fringing capacitance. Moreover, the limitation of distinguishing multi-analyte found in existing studies has remained unresolved.

Herein, we are motivated to propose a novel VOC sensor aimed at addressing the sensor response time and sensitivity through optimization by introducing a porous structure into a PDMS absorption layer and removing the substrate beneath a CSRR structure. The enhanced performance is verified by theoretical analysis, simulation, and experiments involving four common VOCs (i.e., acetone, isopropanol, ethanol, and methanol). Furthermore, two decision-tree-based classification models using different features obtained from the sensor responses are applied to accurate classification of these VOCs. The main scientific contributions of this work are summarized as follows.

1. Employing polymer morphology modification accelerates the swelling phenomena between the polymer and VOC molecules, thereby shortening the sensor's response time. Based on the analysis of fringing electric field and equivalent circuit model, the proposed resonant structure optimization strategy of removing substrate mitigates the inevitable sensitivity loss associated with the porous polymer.
2. Based on the unique absorption characteristics of polymers towards various VOC molecules and their distinct permittivity, a data-driven classification method is proposed utilizing the sensor's transmission response (S_{21}). Furthermore, orthogonal feature parameters are extracted through equivalent circuit analysis, enabling higher accuracy in multi-analyte classification for substances with similar physicochemical properties.

The remaining of this article is organized as follows. Section II describes the operation principle, optimized design, fabrication and the experimental setup. Section III presents the enhanced performance of this sensor by the modified PDMS absorption layer and optimized CSRR structure in VOC detection and classification experiments. Finally, conclusions are drawn in Section IV.

2. Methods

2.1. Working principle of the proposed VOC sensor

The microwave resonator used in this work is based on a microstrip-coupled CSRR structure, which can be excited by an electric field polarized perpendicular to the resonator plane and demonstrates superior performance in dielectric measurements compared to SRR [29]. Fig. 1(a) and (b) show the schematic of the proposed CSRR and its equivalent circuit model, respectively. The CSRR can be described by an LC model, in which C_R and L_R are the total capacitance and inductance of the device, respectively. R represents the equivalent loss through the substrate, and C_C models the coupling capacitance between the microstrip line and the CSRR on the other side. L_L and Z_0 are the inductance

and impedance between the microstrip line and the external ports, respectively. Based on the equivalent circuit, the S_{21} of the CSRR shows a band-notch characteristic at the resonant frequency. Such resonant frequency, also known as the transmission zero frequency, can be described as [30]:

$$f = \frac{1}{2\pi\sqrt{L_R(C_R + C_C)}} \quad (1)$$

When resonance occurs, electromagnetic energy will be stored in the CSRR, resulting in a concentrated fringing electric field in the capacitance region of the CSRR, as depicted in Fig. 1(a). Since the variation in effective permittivity is small within the vicinity of the CSRR when VOC vapors are present, an analyte-sensitive material is required to concentrate VOC molecules within the capacitance region of the CSRR. PDMS is chosen as analyte-sensitive material in this work, which undergoes swelling upon exposure to VOC molecules, resulting in a change in volume. Moreover, VOC molecules with higher permittivity also contribute to an increase in the effective permittivity of PDMS. The changes in the volume of the PDMS layer and permittivity will lead to a more significant variation in f upon absorption of VOC molecules.

Using the perturbation theory [31], the relative change in f can be expressed as:

$$\frac{f - f_0}{f_0} = \frac{\int_{V_M} \Delta\epsilon E_1 \cdot E_0^* dV}{2 \int_{V_C} \epsilon_0 E_0 \cdot E_0^* dV} \quad (2)$$

where f_0 is the initial resonant frequency before the absorption of VOCs. E_0 and E_1 are the fringing electric fields before and after VOC absorption, ϵ_0 is the permittivity of free space, $\Delta\epsilon$ represents the change in permittivity of the PDMS layer due to VOCs absorption, V_M and V_C denote the volumes of the PDMS layer and the electric field, respectively. According to Eq. (2), it can be seen that the swelling of PDMS and the increase of permittivity due to VOC absorption can induce an increase in the resonant frequency. Thus, the concentration of VOC vapor can be quantitatively determined with the sensitivity defined as the change in f (Δf) to the variation in concentration (Δc), which can be given by:

$$S = \frac{\Delta f}{\Delta c} = \frac{|f - f_0|}{c - c_0} \quad (3)$$

However, to qualitatively determine the type of VOC, the amplitude of S_{21} is used as an orthogonal parameter to increase the data dimensionality for the classification of unknown VOC vapors, which will be discussed in detail in the next section. The amplitude of S_{21} can be expressed in terms of R [32] by the following equation:

$$A(\text{in dB}) = 20 \lg \left| \frac{2Z_0}{2Z_0 + R} \right|, \quad (4)$$

where R is related to the loss tangent ($\tan\delta$) of the PDMS layer, which changes upon absorption of VOC with different concentrations and types. Nevertheless, by combining Eq. (1), Eq. (2), and Eq. (4), the qualitative and quantitative VOC detection can be realized by analyzing the changes in the sensor's f and A , as demonstrated in Fig. 1(c).

Fig. 2(a) shows the finite element analysis (FEA) result of the electric field distribution in the capacitance region of the sensor using the High-Frequency Structure Simulator (HFSS) software. It can be seen that the fringing electric field contributes to a capacitance of the resonator (C_R) that includes three components connected in parallel: the electrode-to-electrode capacitance (C_i), the capacitance in the PDMS absorption layer (C_{PDMS}), and the capacitance in the substrate (C_{sub}), i.e., $C_R = C_i + C_{PDMS} + C_{sub}$, as shown in Fig. 2(b). According to Eq. (1), f can be expressed by the following equation when C_{PDMS} is increased by ΔC_{PDMS} when VOC molecules are absorption into the PDMS layer:

$$f = \frac{1}{2\pi\sqrt{L_R(C_i + C_{PDMS} + C_{sub} + C_C + \Delta C_{PDMS})}} \quad (5)$$

Therefore, the Δf in Eq. (3) can be determined as:

$$\Delta f = f_0 - f = \frac{1 - \sqrt{1 + \frac{\Delta C_{PDMS}}{C_i + C_{PDMS} + C_{sub} + C_C}}}{2\pi\sqrt{L_R(C_i + C_{PDMS} + C_{sub} + C_C)}} \quad (6)$$

According to Eqs. (3) and (6), the value of S and Δf are depend on each equivalent capacitance in the sensor equivalent circuit model. Although the porous structure, providing increased active sites for VOC molecules [33], can shorten the sensor's response time, it also cause a reduction in the overall volume of the PDMS absorption layer due to the

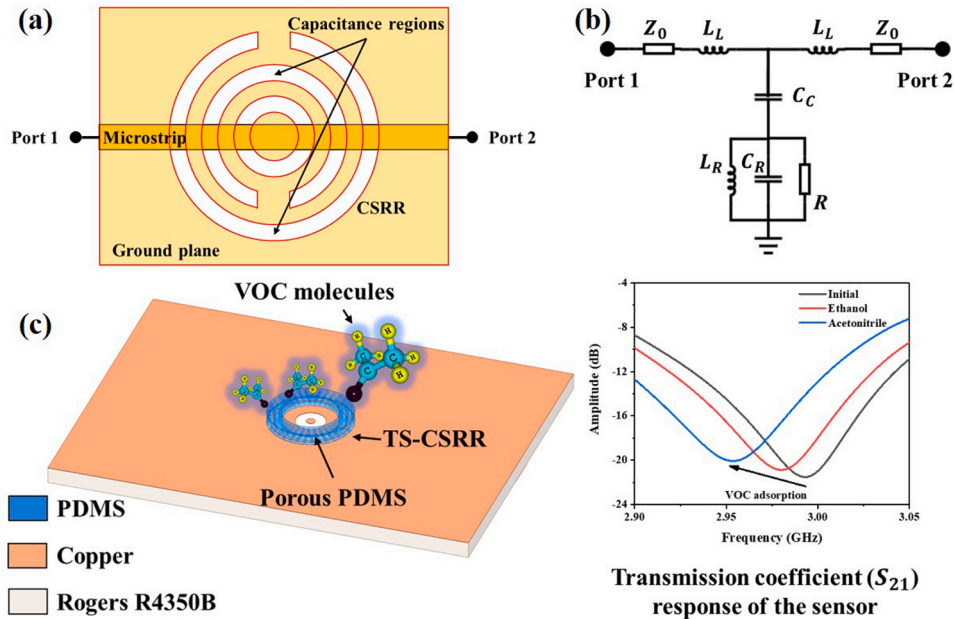


Fig. 1. Structure and working principle of the CSRR-based VOC sensor. (a) Schematic drawing of the microstrip-coupled CSRR in this work. (b) The equivalent circuit model of the CSRR. (c) The working principle of the VOC detection using the proposed sensor. The inset shows changes in f and A due to the absorption of VOC molecules.

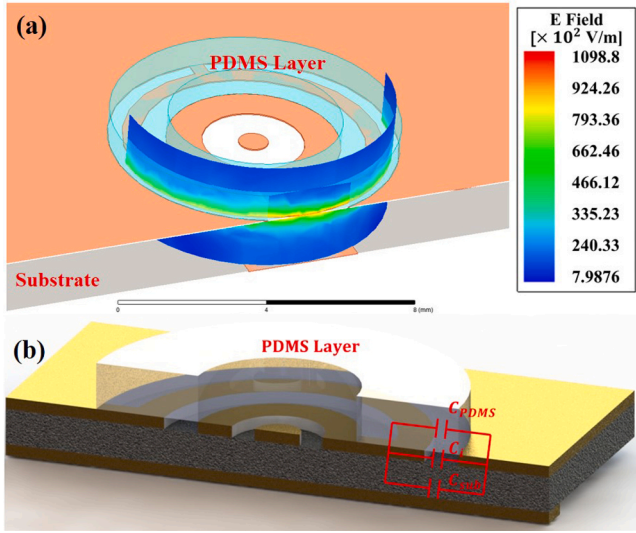


Fig. 2. (a) The FEA simulation result of the electric field distribution in the CSRR. (b) Schematic illustration of the three capacitance components in C_R .

introduction of air, resulting in less VOC absorption and ΔC_{PDMS} . The S and Δf will decrease accordingly. On the other hand, by evaluating the lumped parameter model of C_R shown in Eq. 2(b) and Eq. (6), it can be seen that the S can be increased by reducing C_{sub} . This can be done by removing the substrate material in the capacitance region with a depth higher than the penetration height [34] using the laser grooving technique. In this way, the permittivity in C_{sub} can be minimized to ϵ_0 , leading to a decreased C_{sub} and increased S . Based on this design, we propose a new CSRR structure with improved sensitivity for VOC detection, namely, the trenched-substrate CSRR (TS-CSRR).

The advantage of using TS-CSRR structure is further verified using the FEA method, as shown in Fig. 3, by altering the permittivity, loss

tangent, and thickness of the PDMS absorption layer to simulate the VOC absorption process that induces changes in f and A of the sensor. As shown in Fig. 3(a), f and amplitude of S_{21} both decrease with increasing permittivity in the absorption layer. It should be noted that although $\Delta\epsilon$ only changes f according to Eq. (2), the R in Eq. (4) includes a loss resistance (R_{MUT}) of the PDMS layer, which is related to f and $\tan\delta$ as $R_{MUT} \propto f/\tan\delta$. Thus, the decrease of f would lead to a lower R and consequently reduce the amplitude of S_{21} . When the TS-CSRR structure is used, as shown in Fig. 3(d), Δf and ΔA increase from 0.292 GHz to 0.369 GHz and 1.726 dB to 3.203 dB, respectively. The enhanced responses in f and A can be attributed to the reduced loss and permittivity in the substrate due to substrate removal. In addition, the VOC absorption would also increase the $\tan\delta$ in the PDMS layer, Fig. 3(b) and (e) demonstrate the increased ΔA from 8.544 dB to 10.544 dB by using the TS-CSRR structure under a change of $\tan\delta$ from 0 to 0.1. Fig. 3(c) and (f) evaluate the influence of PDMS swelling on f and A by changing the thickness of the PDMS absorption layer on CSRR and TS-CSRR, respectively. As the layer thickness H increases from 0.6 mm to 1.1 mm, the VOC sensor with TS-CSRR structure shows more significant changes in Δf and ΔA , i.e., 0.029 GHz and 0.23949 dB in TS-CSRR compared to 0.019 GHz and 0.0133 dB in CSRR. Nevertheless, the FEA presented in Fig. 3 well agrees with the analytical analysis in the previous discussion, which supports the sensing performance optimization through substrate removal.

In short, the design of the microwave resonator-based VOC sensor presented in this work is mainly based on a porous absorption layer structure to increase response speed and the structural optimization that utilizes a trenched-substrate structure to increase the sensitivity, realizing a comprehensive improvement of the VOC sensing performance.

2.2. Design and fabrication of the proposed VOC sensor

The proposed microwave VOC sensor is made of a porous PDMS absorption layer loaded on a planar TS-CSRR. The cross-sectional view of the CSRR is schematically shown in Fig. 4(a), which consists of a

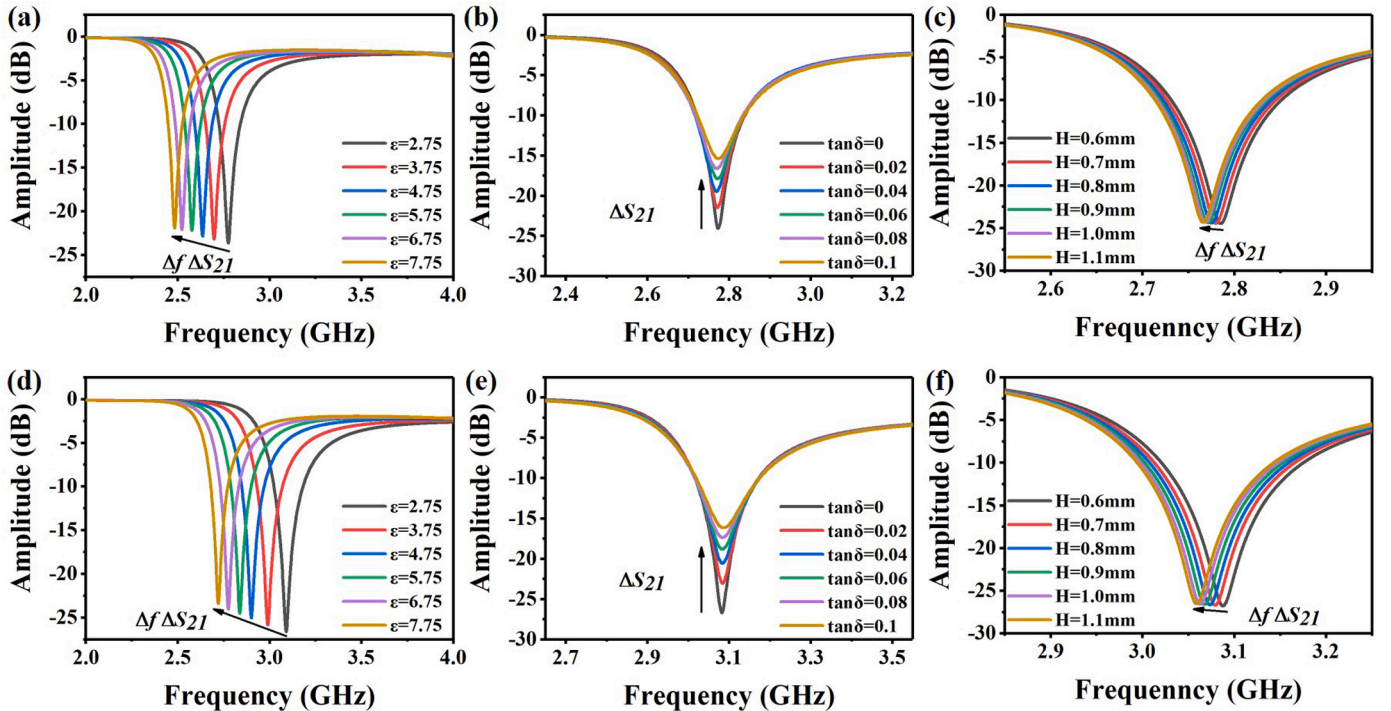


Fig. 3. The FEA simulation results of the resonance response for CSRR and TS-CSRR. (a-c) The influences of permittivity (a), loss tangent (b), and absorption layer thickness (c) on the resonant behaviors of CSRR-based VOC sensor. (d-f) The influences of permittivity (d), loss tangent (e), and absorption layer thickness (f) on the resonant behaviors of TS-CSRR-based VOC sensor.

1.524 mm thick Rogers R4350B substrate layer sandwiched in between a top copper (Cu) ground plane layer and a bottom Cu microstrip layer, with both thickness of 35 μm . The topologies of the bottom and top Cu layers are presented in Fig. 4(b) and (c), respectively. As discussed in the previous section, the detection sensitivity can be improved by reducing C_{sub} . Therefore, trenches were created using the laser grooving technique in the substrate layer from the ground plane in the grooved region, as shown in Fig. 4(c). The height of the trenches was set at 0.8 mm, which was deep enough to cover the penetration depth of the resonator's fringing electric field, thereby minimizing the substrate capacitance. In addition to the CSRR structure, a central ring etched in the ground plane was implemented to reduce C_c between the microstrip and the ground plane, thereby further enhancing sensitivity. To validate the effectiveness of the proposed TS-CSRR in sensitivity enhancement, an unslotted CSRR with the same geometric design parameters was also designed and fabricated for comparison purposes. Fig. 4(d) and (e) show the experimental measured and simulated transmission responses of the CSRR and TS-CSRR. As the C_{sub} is reduced, the initial resonance frequency of the device increases from 3.12 GHz to 3.44 GHz, according to Eq. (5).

Fig. 5(a) shows the simulated electric field distribution over the top plane surface of the TS-CSRR, where the strongest electric field occurs in the inner and outer grooved regions. Therefore, the annular-structured PDMS layer is determined to have a dimension with an outer radius equal to $R_o = 4$ mm and an inner radius equal to $R_i - D_i = 2.4$ mm, as shown in Fig. 4(c), which completely covers the grooved regions of highly-dense electric field. Consequently, any small changes in the dielectric properties in this sensing area induced by the analyte absorption would lead to a noticeable shift in the resonant signal, thus improving the sensing performance.

Fig. 5(b) shows the aluminum alloy mold to fabricate the PDMS

absorption layers used in this work, including solid and porous structures. The fabrication process of the porous PDMS absorption layer is schematically illustrated in Fig. 5(c). To begin with, a mixture of sugar particles with sizes in the range of 250 to 500 μm and erythritol powder was filled into the mold at a mass ratio of 20:1. Next, the mold containing the sugar/erythritol mixture was placed in a convection oven for 2 h at 130°C, which is the temperature between the melting points of erythritol and sugar. As a result, the melted erythritol helps to form an interconnected sugar-based sacrificial mold. After the mold was cooled to room temperature, PDMS solution with base solution (Part A) to curing agent (Part B) ratio of 10:1 (w/w) was drop-casted onto the sugar template and completely cured on a hot plate for 1 h at 90°C. In the next step, warm DI water was used to dissolve the sugar template completely, and the annular-shaped PDMS layer was demolded with the help of a tweezer. The as-fabricated porous PDMS layer was then carefully inspected under the optical microscope to verify no remaining undissolved sugar particles in the PDMS layer. To further characterize the pore structure of the porous PDMS, the scanning electron microscope (SEM) images of its surface and cross-section are presented in Fig. 5(e) and (f), respectively, indicating the presence of numerous pores in the fabricated material. Moreover, it can be seen in Fig. 5(f), these pores inside the PDMS are not isolated but interconnected, thus facilitating the absorption of gas molecules through these pores into the PDMS and contributing to shortened response times. Similarly, the solid PDMS layer with the same dimension was fabricated using the same molding method without using the sugar template for comparison.

After the annular-structured PDMS absorption layer was successfully fabricated, it was aligned with the grooved regions and bonded on top of the TS-CSRR plate using a thin layer of PDMS as the adhesive layer. To investigate the performance of the proposed microwave VOC sensor,

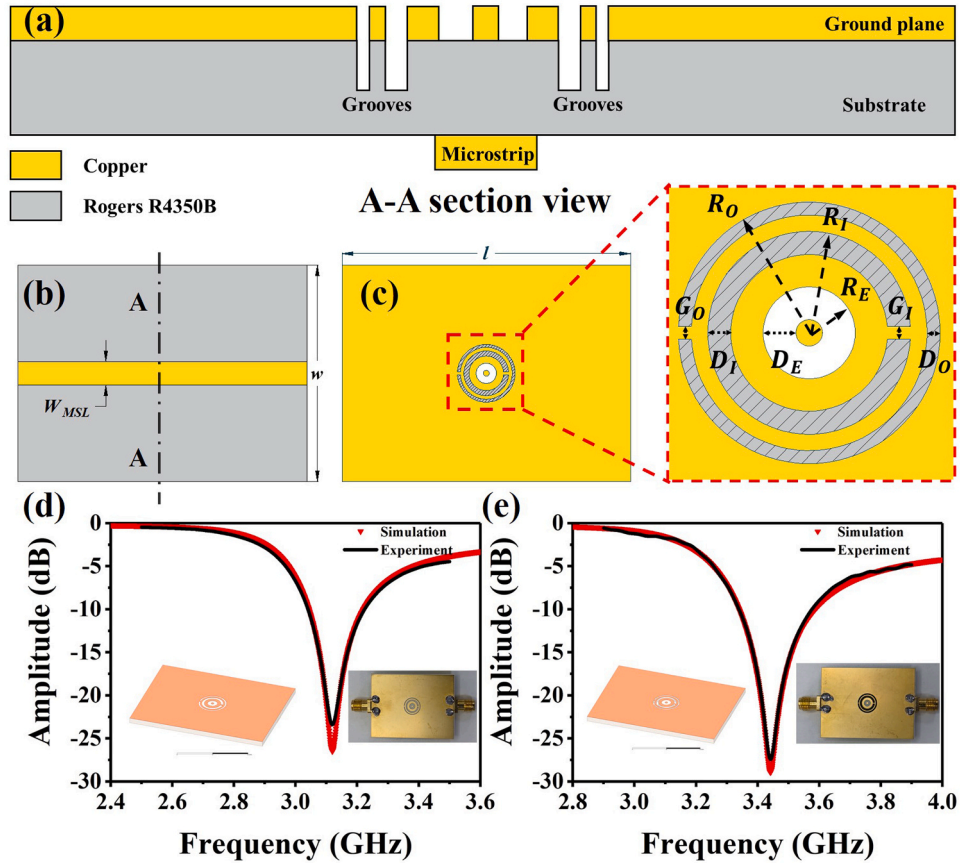


Fig. 4. (a) Cross-sectional schematic of the CSRR. (b) The topology of the bottom Cu layer with a microstrip transmission line structure, where $W_{\text{MSL}} = 3.232\text{mm}$. (c) The topological structure of the top Cu ground plane layer, where the geometrical dimensions are as follows: $l = 40\text{mm}$, $w = 30\text{mm}$, $R_o = 4\text{mm}$, $R_i = 3.1\text{mm}$, $R_E = 1.41\text{mm}$, $D_o = 0.4\text{mm}$, $D_i = 0.7\text{mm}$, $D_E = 1\text{mm}$, $G_o = G_i = 0.4\text{mm}$. The experiment and simulation results of S_{21} responses for (d) CSRR and (e) TS-CSRR.

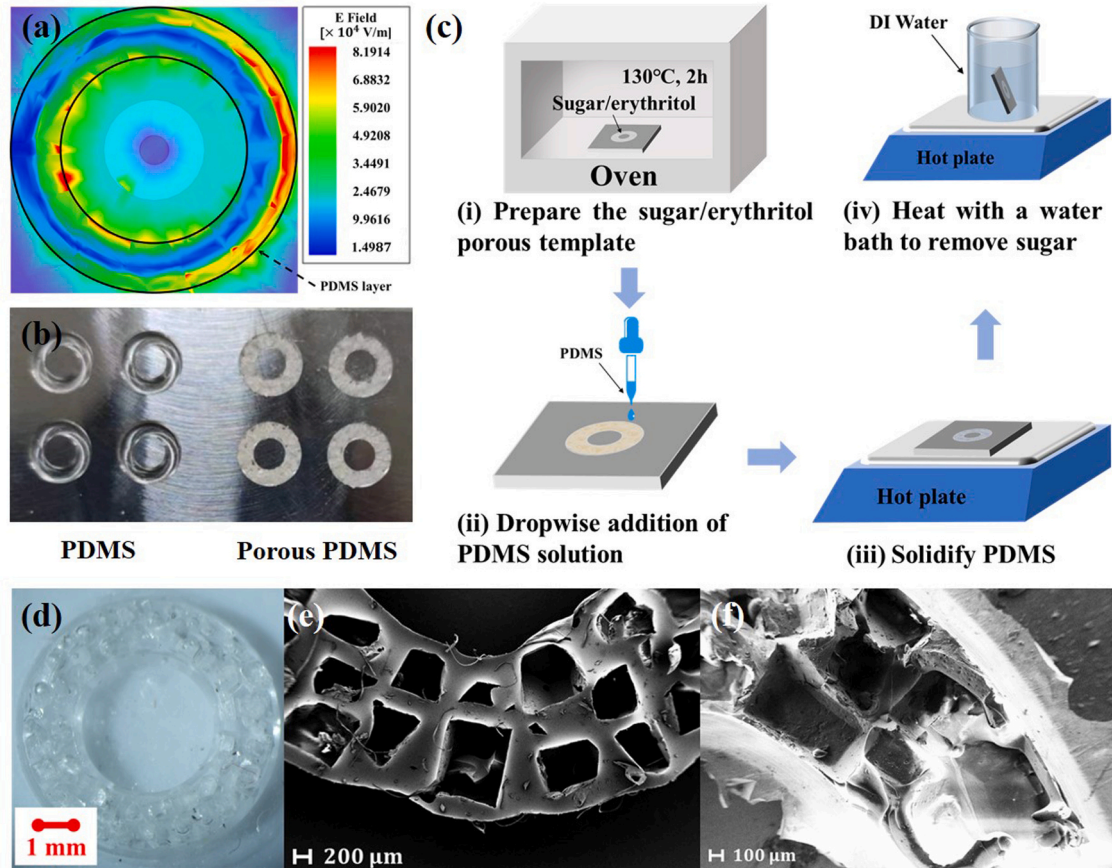


Fig. 5. (a) The simulated electric field distribution over the surface of the proposed TS-CSRR. (b) The aluminum alloy mold used to fabricate the solid and porous PDMS absorption layers. (c) Schematic illustration of the fabrication process of the annular-structured porous PDMS absorption layer. (d) The optical microscope image of the porous PDMS layer. The SEM images of the surface (e) and cross-section (f) of porous PDMS.

three devices were developed with different combinations of absorption layers and planar microwave resonators, as shown in Fig. 6(a), including CSRR with solid PDMS layer, CSRR with porous PDMS layer, and TS-CSRR with porous PDMS layer. It can be seen in Fig. 6(b) that when the solid PDMS layer is integrated on top of the CSRR, the measured f_0 and A exhibit decreases of 0.2198 GHz and 3.85 dB, respectively, due to the increasing dielectric constant and loss in the sensing area. However, more slight decreases of 0.0720 GHz and 1.61 dB can be observed in f_0

and A , respectively, when porous PDMS is used as the absorption layer, which can be attributed to the reduced effective permittivity of porous PDMS due to the air inside the PDMS (dielectric constant and loss both less than PDMS). However, when porous PDMS is paired with TS-CSRR, the measured f_0 and A becomes 0.1868 GHz and 3.95 dB, as shown in Fig. 6(c), which are 159% and 145% larger than the changes caused by porous PDMS layer in CSRR and are comparable to the changes of solid PDMS in CSRR.

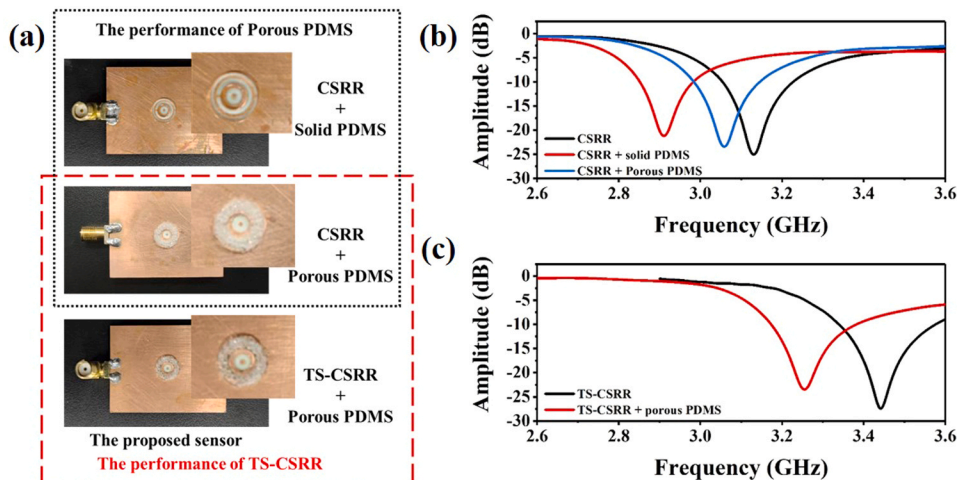


Fig. 6. (a) The three microwave VOC sensors with absorption layers and resonator plates of different structures. (b) The measured resonant response of CSRR, CSRR with solid PDMS layer, and CSRR with porous PDMS layer. (c) The measured resonant response of TS-CSRR, and TS-CSRR with porous PDMS.

2.3. Experimental setup

Fig. 7 illustrates the schematic diagram of the experimental setup for VOC sensing. The testing system includes a customized acrylic testing chamber, a VOC injection micro-syringe, a vector network analyzer (E5063A, Keysight), and a data acquisition system. During the experiment, pure N_2 gas was introduced into the chamber through the gas inlet as both the background gas and the purging gas to provide a high-purity background for sensor characterization and to remove residual gases after each test through the gas outlet, respectively. Four different VOC vapors, including acetone (AC), isopropanol (IPA), ethanol (EtOH), and methanol (MeOH), were introduced into the chamber by injecting their liquid-phase droplets using the micro-syringe located on top of the chamber. With the help of a circulating fan, the VOC liquids can be evaporated to generate a vapor environment with different concentrations (c), which can be determined as:

$$c = \frac{22.4 \times (273 + T) \times \rho \times v \times 10^3}{273 \times V \times M_w} \quad (7)$$

where T is the environment temperature, ρ is the density of VOC in its liquid phase, v is the volume of the injected VOC liquid, M_w is the molar mass of VOC in g/mol, and V is the volume of the chamber [35].

The resonant signal of the VOC sensor was measured at room temperature ($25 \pm 2^\circ\text{C}$) and 1 atm. When analyte vapors were absorbed, the changes in the dielectric constant and the volume of the absorption layer would cause a shift in f and A , which can be recorded using the LabVIEW software at a time interval of 1 s

3. Results and discussions

3.1. Performance of the porous PDMS absorption layer

In this section, the effect of the porous structure on the response time and sensitivity of the PDMS layer is investigated. For this purpose, CSRR-based VOC sensors with solid and porous PDMS absorption layers were fabricated for comparison. Fig. 8 shows the dynamic responses of the VOC sensors in terms of Δf and ΔA upon exposure to AC vapors, with concentrations ranging from 13079 to 65395 ppm. During dynamic gradient concentration tests, pure N_2 gas was first used to evacuate residual gases from the chamber. Then, liquid-phase AC droplets were introduced into the chamber and volatilized into AC vapors that can be

absorbed into the PDMS absorption layer integrated with the TS-CSRR, resulting in a decrease in f and an increase in A until the responses reach a steady state, as indicated in Fig. 8(a) and (b), respectively. Between each consecutive test, pure N_2 gas was used to purge the chamber to allow the sensor responses to return to the baseline level. After that, the sensor is ready to be tested under a different concentration. As shown in Fig. 8(a) and (b), given that the porous structure offers more active sites for VOC molecule absorption, the CSRR with porous PDMS layer responds faster than that of integrated with solid PDMS layer to VOC concentration variation. Nevertheless, it should be noted that there is also a large reduction in Δf and ΔA , which needs to be further improved by the resonator structure optimization.

Furthermore, the sensing performances of solid and porous PDMS-loaded CSRR for four different VOC vapors were further tested. Fig. 9 (a) shows the response time (τ), which characterizes the sensor's response speed, for solid and porous PDMS-loaded CSRR in detecting four different VOC vapors at a concentration of 40000 ppm. The sensor with porous PDMS shows shorter τ values in both f and A for all VOC tests, with τ in f of AC and MeOH as low as 90 s and 54 s, respectively, which is 71.1% and 67.4% lower than that of the sensor with solid PDMS. The response time to different VOC under test follows the order of $\tau_{IPA} > \tau_{EtOH} > \tau_{AC} > \tau_{MeOH}$, which is associated with the different absorption rates for each VOC vapor. The absorption rate of a specific VOC molecule is affected by its molecular diameter [36], characterized based on its relative molecular mass [37], as listed in Table 1. It can be found that except EtOH, the τ for the other three VOCs is proportional to their relative molecular mass. Although EtOH has a relative molecular mass lower than AC, it has the second-highest enthalpy of evaporation among all tested VOCs, also listed in Table 1. Due to this high enthalpy, EtOH requires more energy to evaporate, thus taking a longer time for the EtOH liquid droplet to form the vapor environment. Since this evaporation stage is also included in the response time, this high enthalpy considerably increases τ_{EtOH} . However, although the enthalpy of evaporation for MeOH is higher than that of AC, it has a much smaller relative molecular mass. Therefore, τ_{MeOH} is smaller than the other three VOCs, as observed in the experiment. Moreover, the influence of the porous PDMS structure on the responses of Δf and ΔA is also discussed. As shown in Fig. 9(b), all VOC tests exhibit decreased responses in Δf and ΔA , respectively, for CSRR loaded with a porous PDMS layer, as indicated by the previous analysis. Nevertheless, the selectivity of Δf for each VOC analyte is not changed when the porous PDMS structure is used, as can be seen in Fig. 9(b).

3.2. Performance of the TS-CSRR VOC sensor

For reducing the adverse effect of the porous absorbing layer on sensitivity, TS-CSRR structure is proposed through theoretical and simulation analysis. To evaluate the sensitivity optimization of the proposed TS-CSRR, VOC sensors with two comparative structures, including CSRR and TS-CSRR loaded with porous PDMS absorption layer, were tested in VOC environments of different concentrations. Fig. 10(a) and (b) show $|\Delta f|$ and ΔA responses of two sensors to AC vapors with concentrations ranging from 13,079 to 65,395 ppm. It can be seen that the TS-CSRR exhibits higher responses in terms of both $|\Delta f|$ and ΔA to all tested concentrations for AC. As previously indicated by the theoretical and FEA simulation results, this improved sensitivity can be attributed to the dielectric substrate material removal, which enhances the interaction between the sensing electric field and the materials under test by minimizing the C_{sub} in the fringing capacitance. Besides, since responses in $|\Delta f|$ and ΔA show a highly linear relation with the VOC concentration, a linear regression model is used to fit the experimental data to determine the sensitivity. As seen in Fig. 10(a), the sensitivity of $|\Delta f|$ is determined to be 217.3 Hz/ppm for TS-CSRR in response to AC, respectively, which is 75.5% higher than that of CSRR. Similarly, the A sensitivity of TS-CSRR also exhibits 37.8% improvements for AC, as shown in Fig. 10(b).

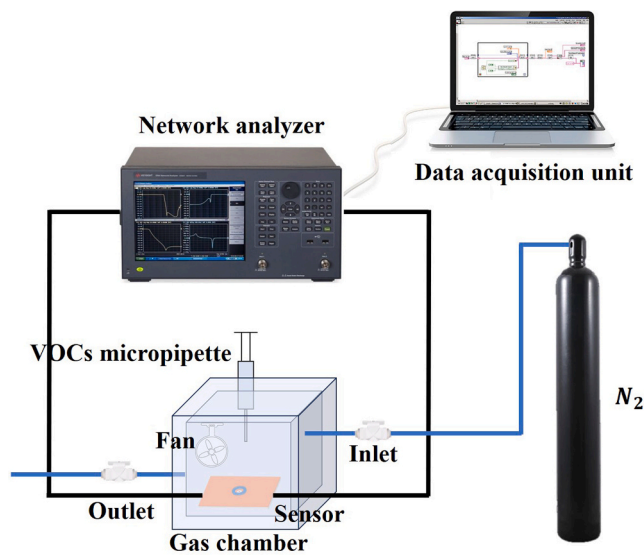


Fig. 7. Schematic illustration of the experimental setup and instrumentation for investigating the response of the proposed VOC sensor to different VOC vapors.

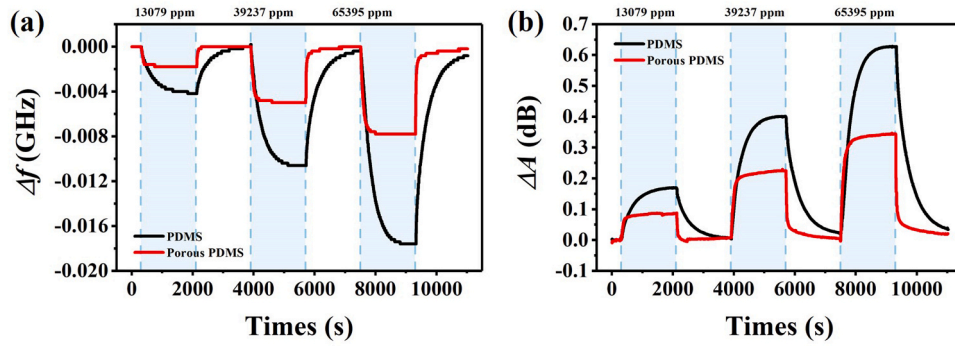


Fig. 8. Dynamic responses of (a) Δf and (b) ΔA of CSRR with solid and porous PDMS layer as a function of time to AC vapors at various steady-state concentrations.

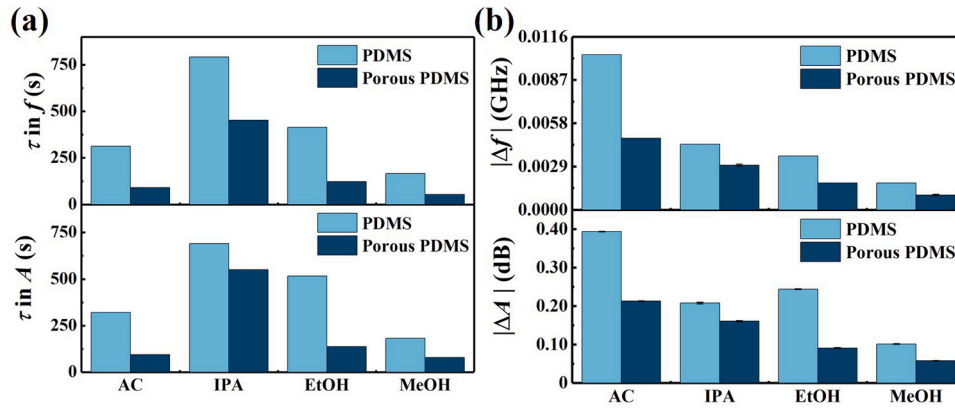


Fig. 9. Comparison of sensing performances for two comparative sensors (CSRR with solid and porous PDMS layer). (a) Response times of two sensors to 4 VOCs at 40,000 ppm. (b) Δf and ΔA responses of two sensors to 4 VOCs at 40,000 ppm.

Table 1
Physical Properties of Vocs Under Test.

	Relative molecular mass	Enthalpy of evaporation (kJ/mol) [38]
AC	58	29.10
IPA	60	39.85
EtOH	46	38.56
MeOH	32	35.21

To further demonstrate the generality of the proposed sensitivity enhancement approach based on substrate removal, four different VOC vapors with a concentration of 40000 ppm were used. As seen in Fig. 11, the sensor with TS-CSRR structure shows higher responses in $|\Delta f|$ and ΔA

for all VOC tests, roughly comparable to CSRR loaded with solid PDMS absorption layer. In summary, the proposed sensor has mitigated the inevitable sensitivity reduction caused by polymer morphology modification through resonant structure optimization, thereby developing a microwave VOC sensor that combines both rapid and sensitive detection of VOCs. It is also worth noting that although the TS-CSRR shows improved sensitivity for both f and A measurements, f is more suitable for quantitative detection of VOC concentration due to its higher stability against ambient noise [39]. Therefore, this work uses f as the VOC concentration measurement parameter, while A is an orthogonal dimension for qualitatively determining VOC types. Nevertheless, the sensing performances in both f and A measurements are discussed for a more comprehensive investigation of the proposed optimization

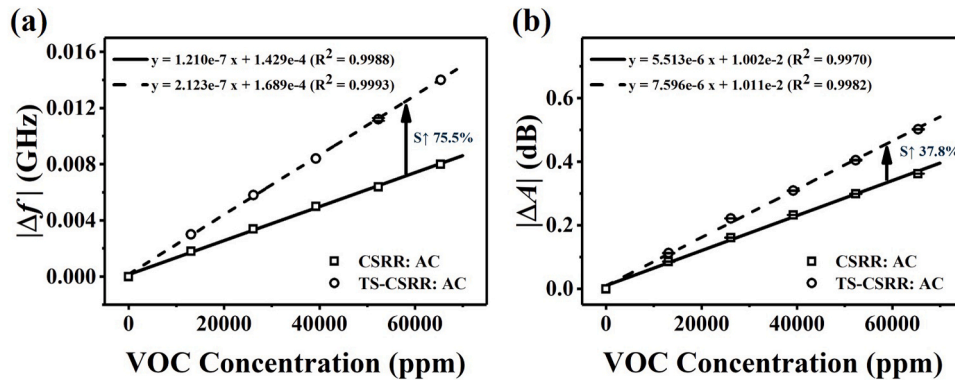


Fig. 10. The sensing performances of the proposed TS-CSRR VOC sensor compared with the CSRR structure for detecting AC vapors with different concentrations. Experimental results of (a) $|\Delta f|$ and (b) ΔA are shown as a function of VOC concentration.

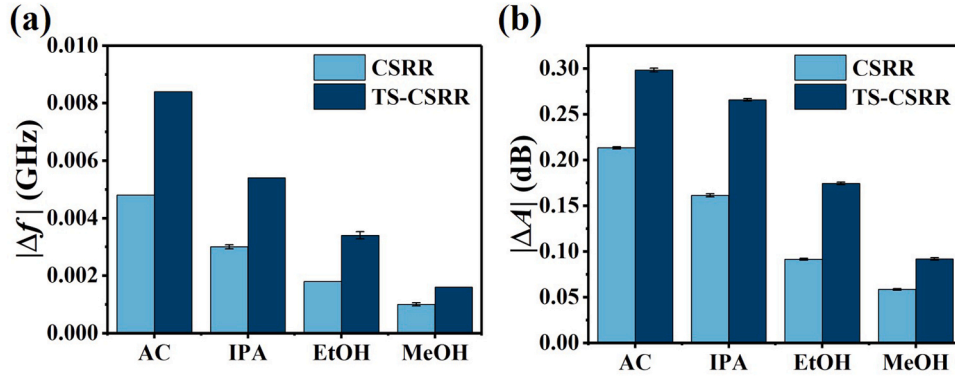


Fig. 11. (a) Δf and (b) ΔA responses of sensors with CSRR and TS-CSRR structures to four different VOC vapors at a concentration of 40000 ppm.

method.

3.3. Performance of TS-CSRR with porous PDMS layer in VOCs detection

To investigate the sensing performances of the TS-CSRR loaded with porous PDMS absorption layer, experimental tests were conducted for four VOC vapors with a wide concentration range of up to 60,000 ppm, as shown in Fig. 12. The real-time responses of f at the lowest concentration are presented in the insert diagrams. It can be seen that the $|\Delta f|$ to each VOC vapor increases linearly with the concentration. Thus, a linear regression model can fit the measured data as a function of VOC concentration. The sensor responses show high linearity for four VOCs with a low mean relative error (MRE) value in the range of 1.07% to 9.82% and a high R^2 value of more than 0.98, indicating the accurate measurement of VOC concentration using the proposed sensor with the linear regression model in a wide concentration range. In addition, the limit of detection (LOD) was also investigated for the proposed VOC sensor. Based on the general criterion for the minimum detectable signal, which requires a signal-to-noise ratio greater than 3 [40], the LOD is determined to be 295 ppm, 354 ppm, 613 ppm, and 1289 ppm for AC, IPA, EtOH, and MeOH, respectively, as indicated in Fig. 12(a-d).

Comparing the sensor's sensitivity to four VOCs, as seen in Fig. 12(a-d), the maximum sensitivity is 217 Hz/ppm in AC detection. The sensitivity for the other three VOC vapors also shows a high value of 181 Hz/ppm, 104 Hz/ppm, and 50 Hz/ppm for IPA, EtOH, and MeOH,

respectively. According to Eq. (2), the absorption of VOC molecules causes a change in f due to the increase in thickness and permittivity of the PDMS absorption layer. Table 2 presents the swelling ratio of the PDMS upon exposure to different VOCs, indicating the significance of thickness change in PDMS due to absorption. According to Maxwell Garnett equation [41], the permittivity of the PDMS can be expressed as follows:

$$\epsilon_{eff} = \epsilon_{PDMS} + 3h\epsilon_{PDMS} \frac{\epsilon_{VOC} - \epsilon_{PDMS}}{\epsilon_{VOC} + 2\epsilon_{PDMS} - h(\epsilon_{VOC} - \epsilon_{PDMS})}, \quad (8)$$

where ϵ_{eff} represents the equivalent permittivity of the PDMS after absorption of VOC molecules, ϵ_{PDMS} and ϵ_{VOC} represent the permittivity of PDMS and VOC, respectively, and h is the volume fraction of the absorbed molecules, which can be approximated by the weight change of PDMS after absorption of VOC molecules and their densities. Table 2 shows the physical properties of each VOC analyte and PDMS. Consequently, due to changes in ϵ_{eff} and different swelling ratios of the PDMS absorption layer to each VOC vapor, the sensor demonstrates different sensitivities to the four tested VOCs, i.e., higher swelling ratio and higher ϵ_{eff} lead to higher responses in Δf . Nevertheless, the proposed VOC sensor using TS-CSRR loaded with porous PDMS absorption layer has been demonstrated to detect VOC vapors with high sensitivity, low LOD, and wide linear detection range. Table 3 shows the comparison of the performance of the proposed sensor and those of recent microwave VOC sensors [42-44,18,45]. By comparison, the proposed sensor

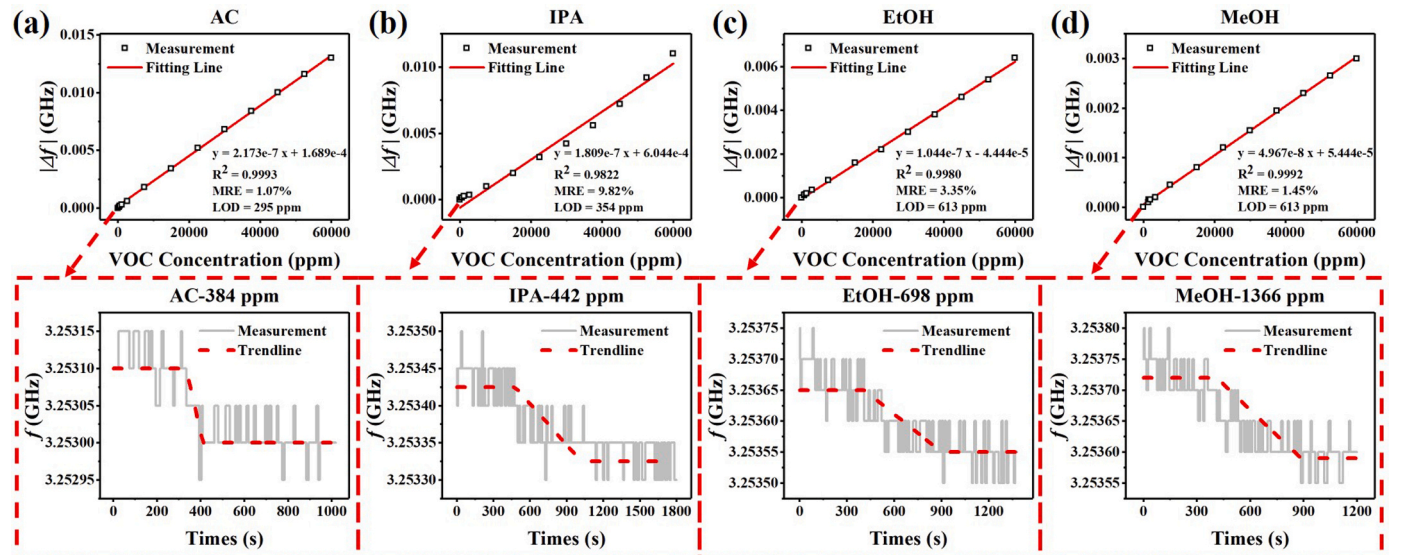


Fig. 12. Sensing performances of the proposed VOC sensor with TS-CSRR structure loaded with porous PDMS in response to different VOC vapors: (a)AC; (b)IPA; (c) EtOH; (d)MeOH.

Table 2
Physical Properties of Voc Analyte And Pdms.

Materials	Swelling ratio of PDMS [12]	Weight change ratio [12]	ϵ [49,50]	Density (gml ⁻¹) [38,51]	h (Calculated)	ϵ_{eff} (Calculated)
AC	1.31	1.33	21	0.790	0.399	5.68
IPA	1.26	1.23	20.3	0.785	0.279	4.60
EtOH	1.11	1.08	24.6	0.790	0.098	3.43
MeOH	1.03	1.04	32.7	0.791	0.051	3.07
PDMS	-	-	2.75	0.965	-	-

exhibits a significantly higher sensitivity compared to other polymer-based sensors [44,45], while exhibiting a larger detection range compared to sensors used nanomaterials [42,43]. It should be noted that although the sensitivity of this work is lower than that of another microwave resonator [18] incorporating porous PDMS, which is primarily attributed to differences in operating frequency (2.7 times that of the proposed sensor) and PDMS volume (6.7 times that of the proposed sensor), the proposed sensor boasts a significantly broader linear detection range. Besides, the proposed resonant structure optimization approach based on fringing electric field and equivalent circuit model has the potential to further improve the performance of various planar microwave-based sensors, such as SRR [46], coplanar waveguide resonator [47], electric inductor-capacitor resonator [48], etc. Furthermore, the multi-analyte classification achieved based on the varying impacts of VOCs' intrinsic permittivity and their interactions with the sensitive material on orthogonal parameters extracted from the sensor transmission response, is another primary feature absent in the aforementioned papers.

3.4. Recognition of VOCs: comparison of different classification models

Although the sensor exhibits different sensitivities in response to distinct VOCs, it's insufficient to distinguish different VOC vapors. For instance, 15,000 ppm AC and 22,500 ppm IPA would induce a similar Δf of 3.4 MHz and 3.2 MHz, respectively. To qualitatively determine types of VOCs at various concentrations, A , determined by $\tan\delta$ of the absorption layer, can be used as an independent orthogonal parameter to unveil the cross-sensitive of different VOCs together with f . Besides, according to Eq. (1) and Eq. (4), the variations in f and A are manifested in the equivalent circuit model by evaluating changes in C_R and R . Therefore, two classification models, M1 and M2, for recognizing different VOCs are developed based on the above two pairs of orthogonal parameters, i.e., (f , A) and (C_R , R), respectively, and results are compared.

The lumped parameters C_R and R can be extracted by fitting the experimentally measured A into its theoretical expression developed from the equivalent circuit model, which can be expressed as:

$$A \text{ (in dB)} = 20\lg \left| \frac{2Z_0 Z_{CSRR}}{Z_1 Z_{CSRR} + Z_2 Z_{CSRR} + Z_1 Z_2} \right|, \quad (9)$$

where Z_{CSRR} represents the resonator impedance. Z_1 and Z_2 denote the impedances of the microstrip line coupled with Port 1 and Port 2, respectively, as shown in Fig. 13(a). The result of the curve fitting is then presented in Fig. 13(b) based on the experimentally obtained initial values using lumped parameters determination method described in [30].

Table 3
Comparison of Different Microwave Voc Sensors.

Ref.	Sens. Method	Sensitive material	Operating frequency	Detection range	Sensitivity	Multi-analyte classification
[44]	CPWs	PGMA/POEGMA copolymer	9.29 GHz	~1600 to 16,000 ppm	104 Hz/ppm (AC)	No
[45]	Differential SRR	PDMS	~5 GHz	3878 to 62,048 ppm	194 Hz/ppm (AC)	No
[42]	Differential Resonator	PEDOT:PSS-MWCNTs	0.65 GHz	500 to 2000 ppm	643 Hz/ppm (EtOH)	No
[43]	Microstrip feedline	MXene	1.834 GHz	8000 to 80,000 ppm	700 Hz/ppm (AC)	No
[18]	SRR	Porous PDMS	8.63 GHz	5000 to 25,000 ppm	2640 Hz/ppm (AC)	No
This work	TS-CSRR	Porous PDMS	3.25 GHz	295 to 60,000 ppm	217 Hz/ppm (AC)	Yes

Fig. 14(a) and (b) depict the sensor's measurements of four VOCs under eight concentrations, ranging from 7500 ppm to 60,000 ppm, in 3D plots, showing the relative change rate of (f , A) and (C_R , R). It can be seen that each VOC analyte occupies a distinct space, demonstrating the potential of recognizing different VOC vapors using both pairs of parameters. Thus, 121 sensor response data were recorded at each concentration, with 3872 samples in total. The orthogonal parameters mentioned above are then selected as inputs to establish two classification models based on the decision tree model. To provide a more comprehensive and reliable evaluation of the sensing performance, especially for a relatively small data set, 10-fold cross-validation is used by dividing the dataset into ten subsets, each subset serving as a validation set. The classification results for recognizing VOCs using two different models are presented in Fig. 15. Each row in the confusion matrix represents the true class, and each column represents the predicted class. The numbers in the grid show the quantity of samples predicted by the model. It can be seen that M1 and M2 both demonstrate high overall accuracies of 98.71% and 99.59%, respectively. Though both models show excellent recognition performances, M2 provides a higher classification accuracy, especially for distinguishing alcoholic analytes. Thus, by using M2 instead of M1, the misclassification of IPA, EtOH, and MeOH has been reduced from 50 to 16, significantly improving the recognition performance. As seen in Fig. 14(a), the measured data in the magnified projection diagram indicates a close response in the lower concentration range. However, the difference in the C_R and R constructed space is more pronounced, as shown in Fig. 14 (b), contributing to a more accurate recognition result. Comparing these two models, directly obtained f and A from experimental measurement is highly dependent on the instrument precision since only the data point at the peak of the resonance signal is used. On the other hand, extracting C_R and R based on the equivalent circuit model utilizes all S_{21} data in the full sweep frequency range, significantly reducing the error. Thus, the model M2 based on the curve fitting method can reveal more dielectric details among analytes with similar chemical properties, which is reflected in more divergent data plots, as presented in the projection diagram in Fig. 14(b). In short, two recognition models based on transmission response and equivalent circuit can achieve high-accuracy classification of VOCs, with the equivalent circuit method demonstrating superior performance in distinguishing substances with similar properties.

4. Conclusion

A novel VOC sensor, integrating a TS-CSRR structure with a porous PDMS absorption layer, has been developed here for fast and sensitive concentration detection of different VOCs and qualitative classification

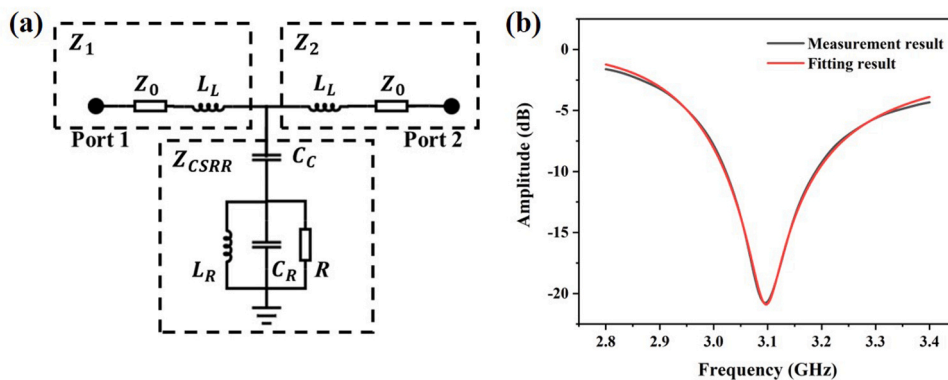


Fig. 13. (a) The equivalent circuit model of the proposed sensor and impedance of each branch. (b) Fitting result of the sensor's transmission response.

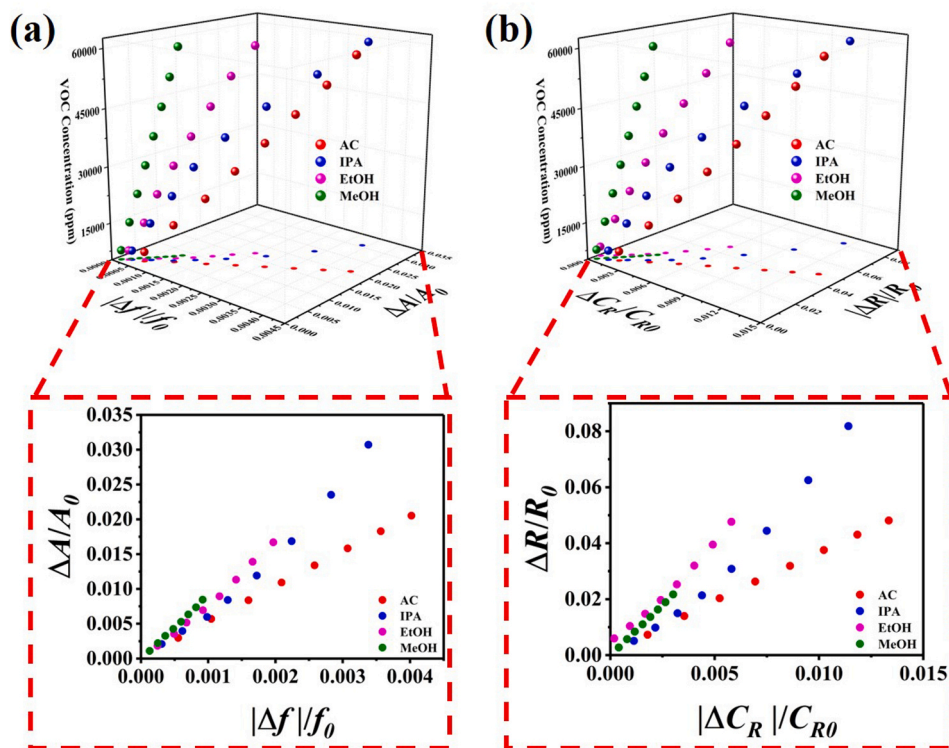


Fig. 14. Distribution of sensor response in the feature spaces and projections. (a) The relative change rate of f and A . (b) The relative change rate of C_R and R .

of multiple VOCs. The introduced porous structure greatly enlarges the specific surface area of the PDMS absorption layer and thus promote the interaction between absorbed analytes and the absorption layer. In this way, the response time is significantly shortened, up to 71.1%. A maximum enhancement of 75.5% in the frequency sensitivity and 37.8% in the amplitude sensitivity are reached in experiments by constructing this TS-CSRR structure to eliminate the effect by the equivalent through-substrate capacitor coupled in the fringing capacitance, which mitigates the inevitable sensitivity reduction caused by polymer morphology modification. The proposed sensor demonstrates outstanding performance in measuring the concentration of various VOCs, among which a highest sensitivity of 217 Hz/ppm and a lowest LOD of 295 ppm are achieved for AC. Two decision-tree-based classification models based on measured resonant parameters and lumped parameters extracted from the responses given by our sensor are built and correctly distinguish multiple VOCs with high overall accuracies of 98.71% and 99.59%, respectively. Moreover, the latter one with extracted lumped parameters, which fully utilizes the generated responses throughout the full

swept frequency band, reduces the error caused by limited information obtained at the resonant frequency in the former classification model. By this means, superior performance is achieved by the latter model when classifying VOCs like IPA, EtOH, and MeOH that share similar physicochemical properties. Therefore, the proposed sensor technology has made an important contribution to the performance enhancement of microwave VOC sensor from comprehensive perspectives of absorption material modification, the resonant structure optimization, and an effective feature extraction method for accurate data-driven VOC classification, achieving an optimized balance between response time and sensitivity.

Environmental implication

Volatilorganic compounds (VOCs) are prevalent environmental pollutants in industrial processes, posing significant threats to both the environment and human health. The severity of VOCs' hazards depends on their types and concentrations. For instance, acetone has threshold

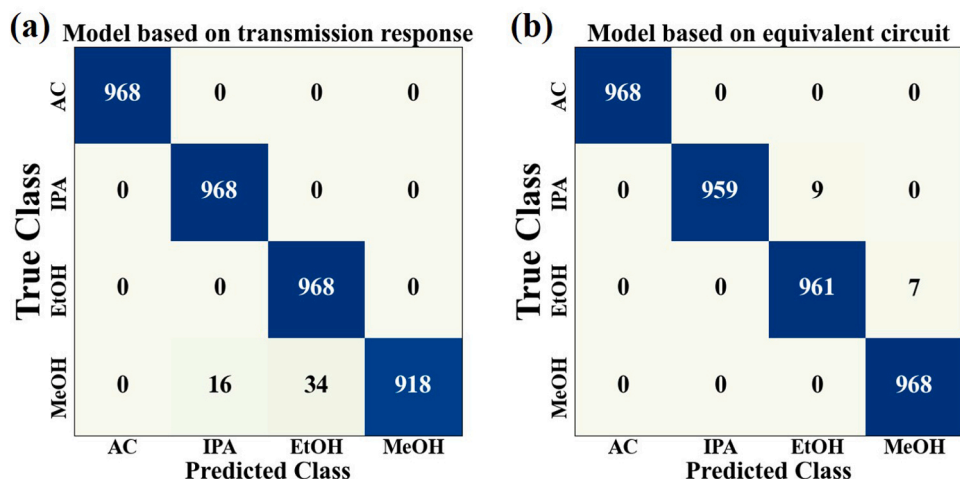


Fig. 15. 10-fold cross-validation results: (a) model M1 based on f and A ; (b) model M2 based on C_R and R .

limit and immediately dangerous to life or health values of 750 ppm and 20000 ppm, respectively, while isopropanol's corresponding values are 400 ppm and 25,000 ppm. This work introduces a novel approach to VOC detection and classification at room temperature, with the advantage of a broad linear detection range spanning from ppm.

CRedit authorship contribution statement

Chenyang Gao: Writing – review & editing. **Shuyu Fan:** Writing – review & editing, Validation, Methodology. **Mengyao Fu:** Writing – review & editing, Investigation, Conceptualization. **Zhiqiang Ma:** Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Dibo Hou:** Supervision, Project administration, Funding acquisition. **Yunqi Cao:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Haozhen Chi:** Visualization, Conceptualization. **Wei Li:** Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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References

- [1] Kang, S., Zhao, K., Yu, D.-G., Zheng, X., Huang, C., 2022. Advances in biosensing and environmental monitoring based on electrospun nanofibers. *Adv Fiber Mater* 4 (3), 404–435. <https://doi.org/10.1007/s42765-021-00129-0>.
- [2] Li, A.J., Pal, V.K., Kannan, K., 2021. A review of environmental occurrence, toxicity, biotransformation and biomonitoring of volatile organic compounds. *Environ Chem Ecotoxicol* 3, 91–116. <https://doi.org/10.1016/j.enceco.2021.01.001>.
- [3] Rajabi, H., Hadi Mosleh, M., Mandal, P., Lea-Langton, A., Sedighi, M., 2020. Emissions of volatile organic compounds from crude oil processing - Global emission inventory and environmental release. *Sci Total Environ* 727, 138654. <https://doi.org/10.1016/j.scitotenv.2020.138654>.
- [4] Yang, S., Perret, V., Hager, J., Niculita-Hirzel, H., Goyette, P., J., Licina, D., 2020. Volatile organic compounds in 169 energy-efficient dwellings in Switzerland. *Indoor Air* 30 (3), 481–491. <https://doi.org/10.1111/ina.12667>.
- [5] Yin, F., Yue, W., Li, Y., Gao, S., Zhang, C., Kan, H., et al., 2021. Carbon-based nanomaterials for the detection of volatile organic compounds: a review. *Carbon* 180, 274–297. <https://doi.org/10.1016/j.carbon.2021.04.080>.
- [6] Beria, Y., Das, G.S., Buragohain, A., Chamuah, B.B., 2023. Highly sensitive miniaturized octagonal DS-CSRR sensor for permittivity measurement of liquid samples. *IEEE Trans Instrum Meas*. <https://doi.org/10.1109/TIM.2023.3307172>.
- [7] Niksan, O., Jain, M.C., Shah, A., Zarifi, M.H., 2022. A nonintrusive flow rate sensor based on microwave split-ring resonators and thermal modulation. *Ieee Trans Microw Theory Tech* 70 (3), 1954–1963. <https://doi.org/10.1109/TMTT.2022.3142038>.
- [8] Mohammadi, S., Adhikari, K.K., Jain, M.C., Zarifi, M.H., 2021. High-resolution, sensitivity-enhanced active resonator sensor using substrate-embedded channel for characterizing low-concentration liquid mixtures. *IEEE Trans Microw Theory Tech* 70 (1), 576–586. <https://doi.org/10.1109/TMTT.2021.3109599>.
- [9] Abbasi, Z., Shariaty, P., Nosrati, M., Hashisho, Z., Daneshmand, M., 2019. Dual-band microwave circuits for selective binary gas sensing system. *IEEE Trans Microw Theory Tech* 67 (10), 4206–4219. <https://doi.org/10.1109/TMTT.2019.2934459>.
- [10] Sumdani, M.G., Islam, M.R., Yahaya, A.N.A., Safie, S.I., 2022. Recent advancements in synthesis, properties, and applications of conductive polymers for electrochemical energy storage devices: a review. *Polym Eng Sci* 62 (2), 269–303. <https://doi.org/10.1002/pen.25859>.
- [11] Li, X., Zhang, L., Yang, Z., He, Z., Wang, P., Yan, Y., et al., 2020. Hydrophobic modified activated carbon using PDMS for the adsorption of VOCs in humid condition. *Sep Purif Technol* 239, 116517. <https://doi.org/10.1016/j.seppur.2020.116517>.
- [12] Rumens, C.V., Ziai, M.A., Belsey, K.E., Batchelor, J.C., Holder, S.J., 2015. Swelling of PDMS networks in solvent vapours; applications for passive RFID wireless sensors. *J Mater Chem C* 3 (39), 10091–10098. <https://doi.org/10.1039/c5tc01927c>.
- [13] McGinn, C.K., Lampert, Z.A., Kyminis, I.J., 2020. Review of gravimetric sensing of volatile organic compounds, 5 (6), 1514–1534.
- [14] Acharyya, S., Nag, S., Kimbature, S., Ghose, A., Pal, A., Guha, P.K.J., 2021. Selective discrimination of VOCs applying gas sensing kinetic analysis over a metal oxide-based chemiresistive gas sensor, 6 (6), 2218–2224.
- [15] Ali, F.I., Mahmoud, S.T., Awwad, F., Greish, Y.E., Abu-Hani, A.F.J.C., 2020. Low power consumption and fast response H2S gas sensor based on a chitosan-CuO hybrid nanocomposite thin film. p. 236, 116064.
- [16] Viespe, C., Miu, D.J.S., 2018. Characteristics of surface acoustic wave sensors with nanoparticles embedded in polymer sensitive layers for VOC detection, 18 (7), 2401.
- [17] Islam, M.Z., Fu, Y., Deb, H., Hasan, M.K., Dong, Y., Shi, S.J.E.P.J., 2023. Polymer-based low dielectric constant and loss materials for high-speed communication network: Dielectric constants and challenges. 200, p. 112543.
- [18] Mirzaei, H., McClelland, J., Sharma, D., Arjmand, M., Zarifi, M.H.J.A.A.M., Interfaces, 2023. A Microwave Voyage into Swelling Phenomenon: Investigation of Polydimethylsiloxane and VOCs Interaction. 15(31), p. 38008–38017.
- [19] Liu, W., Yang, X., Niu, Y., Sun, H., 2019. Improve planar multiple split-ring sensor for microwave detection applications. *Sens Actuators A Phys* 297, 111542. <https://doi.org/10.1016/j.sna.2019.111542>.
- [20] Fu, J.-H., Wu, W.-J., Wang, D.-W., Zhao, W.-S.J.I.S.J., 2022. High-sensitivity microfluidic sensor based on quarter-mode interdigitated spoof plasmons. 22(24), p. 23888–23895.

- [21] Govind, G., Akhtar, M.J., 2019. Metamaterial-inspired microwave microfluidic sensor for glucose monitoring in aqueous solutions. *IEEE Sens J* 19 (24), 11900–11907. <https://doi.org/10.1109/JSEN.2019.2938853>.
- [22] Ebrahimi, A., Scott, J., Ghorbani, K., 2019. Ultrahigh-sensitivity microwave sensor for microfluidic complex permittivity measurement. *IEEE Trans Microw Theory Tech* 67 (10), 4269–4277. <https://doi.org/10.1109/TMTT.2019.2932737>.
- [23] A. Kandwal Z. Nie T. Igbe J. Li Y. Liu L.W. Liu et al. Measurement, 2020. Surface plasmonic feature microwave sensor with highly confined fields for aqueous-glucose and blood-glucose measurements. 70, p. 1–9.
- [24] Alotaibi, S.A., Cui, Y., Tentzeris, M.M., 2019. CSRR based sensors for relative permittivity measurement with improved and uniform sensitivity throughout [0.9–10.9] GHz band. *IEEE Sens J* 20 (9), 4667–4678. <https://doi.org/10.1109/JSEN.2019.2951172>.
- [25] Su, L., Muoz-Enano, J., Vlez, P., Gil-Barba, M., Casacuberta, P., Martin, F., 2021. Highly sensitive reflective-mode phase-variation permittivity sensor based on a coplanar waveguide terminated with an open complementary split ring resonator (OCSRR). *IEEE Access* 9, 27928–27944. <https://doi.org/10.1109/ACCESS.2021.3058575>.
- [26] Xu, H., Wu, W.-J., Zhao, W.-S., Tang, W., Sun, L.J.I.To.I., 2024. Ultrasensitive edible oil sensor based on spiral SLSF combined with stepped structure, 73, 1–9.
- [27] Govind, G., Akhtar, M.J., 2020. Design of an ELC resonator-based reusable RF microfluidic sensor for blood glucose estimation. *Sci Rep* 10 (1). <https://doi.org/10.1038/s41598-020-75716-z>.
- [28] Albishi, A.M., Ramahi, O.M., 2018. Highly sensitive microwaves sensors for fluid concentration measurements. *Ieee Microw Wirel Compon Lett* 28 (4), 287–289. <https://doi.org/10.1109/LMWC.2018.2805866>.
- [29] Boybay, M.S., Ramahi, O.M., 2012. Material characterization using complementary split-ring resonators. *IEEE Trans Instrum Meas* 61 (11), 3039–3046. <https://doi.org/10.1109/TIM.2012.2203450>.
- [30] Bonache, J., Gil, M., Gil, I., Garca-Garca, J., Martn, F., 2006. On the electrical characteristics of complementary metamaterial resonators. *IEEE Microw Wirel Compon Lett* 16 (10), 543–545. <https://doi.org/10.1109/LMWC.2006.882400>.
- [31] Singh, S.K., Tiwari, N.K., Yadav, A.K., Akhtar, M.J., Kar, K.K., 2019. Design of ZnO/N-doped graphene nanohybrid incorporated RF complementary split ring resonator sensor for ammonia gas detection. *IEEE Sens J* 19 (18), 7968–7975. <https://doi.org/10.1109/JSEN.2019.2918304>.
- [32] Lee, C.-S., Yang, C.-L., 2014. Complementary split-ring resonators for measuring dielectric constants and loss tangents. *IEEE Microw Wirel Compon Lett* 24 (8), 563–565. <https://doi.org/10.1109/LMWC.2014.2318900>.
- [33] Peng, X., Chen, L., Liu, S., Hu, L., Zhang, J., Wang, A., et al., 2021. Insights into the interfacial interaction mechanisms of p-arsanilic acid adsorption on ionic liquid modified porous cellulose. *J Environ Chem Eng* 9 (3), 105225. <https://doi.org/10.1016/j.jece.2021.105225>.
- [34] Bhuiyan, R.H., Alam, N., Caicedo, J.M., Ali, M., 2015. Real-time wireless moisture sensing in concrete using interdigitated stick-on Sensors. *Sens Transducers* 195, 30–38.
- [35] Ihdene, Z., Mekki, A., Mettai, B., Mahmoud, R., Hamada, B., Chehimi, M.M., 2014. Quartz crystal microbalance VOCs sensor based on dip coated polyaniline emeraldine salt thin films. *Sens Actuators B: Chem* 203, 647–654. <https://doi.org/10.1016/j.snb.2014.07.030>.
- [36] Li, X., Zhang, L., Yang, Z., Wang, P., Yan, Y., Ran, J., 2020. Adsorption materials for volatile organic compounds (VOCs) and the key factors for VOCs adsorption process: a review. *Sep Purif Technol* 235, 116213. <https://doi.org/10.1016/j.seppur.2019.116213>.
- [37] Schrder, E., Mller, G., Arndt, K.-F., 2022. Polymer characterization. Walter de Gruyter GmbH & Co KG.
- [38] Hutchinson, J.P., Remenyi, T., Nesterenko, P., Farrell, W., Groeber, E., Szucs, R., et al., 2012. Investigation of polar organic solvents compatible with Corona Charged Aerosol Detection and their use for the determination of sugars by hydrophilic interaction liquid chromatography. *Anal Chim Acta* 750, 199–206. <https://doi.org/10.1016/j.aca.2012.04.002>.
- [39] Zhang, L., Yang, H., Tan, Q., Jing, L., Zhang, W., Xiong, J., et al., 2021. Wireless detection of biogenic amines using a split-ring resonator with silver nanoparticles-decorated molybdenum disulfide. *Sens Actuators B Chem* 343, 130155. <https://doi.org/10.1016/j.snb.2021.130155>.
- [40] Bao, Y., Xu, P., Cai, S., Yu, H., Li, X., 2018. Detection of volatile-organic-compounds (VOCs) in solution using cantilever-based gas sensors. *Talanta* 182, 148–155. <https://doi.org/10.1016/j.talanta.2018.01.086>.
- [41] Zarifi, M.H., Sohrabi, A., Shaibani, P.M., Daneshmand, M., Thundat, T., 2014. Detection of volatile organic compounds using microwave sensors. *IEEE Sens J* 15 (1), 248–254. <https://doi.org/10.1109/JSEN.2014.2345477>.
- [42] Bahoumina, P., Hallil, H., Lachaud, J.-L., Abdelghani, A., Frigui, K., Bila, S., et al., 2017. Microwave flexible gas sensor based on polymer multi wall carbon nanotubes sensitive layer (others). *Sens Actuators B: Chem* 249, 708–714. <https://doi.org/10.1016/j.snb.2017.04.127>.
- [43] Khorsand Kazemi, K., Hu, S., Niksan, O., Adhikari, K.K., Tanguy, N.R., Li, S., et al., 2022. Low-profile planar antenna sensor based on Ti3C2Tx MXene membrane for VOC and humidity monitoring. *Adv Mater Interfaces* 9 (13), 2102411. <https://doi.org/10.1002/admi.202102411>.
- [44] Li, H., Chen, Z., Borodinov, N., Shao, Y., Luzinov, I., Yu, G., et al., 2017. Multi-frequency measurement of volatile organic compounds with a radio-frequency interferometer. *IEEE Sens J* 17 (11), 3323–3331. <https://doi.org/10.1109/JSEN.2017.2692521>.
- [45] Mohammadi, S., Zarifi, M.H., 2020. Differential microwave resonator sensor for real-time monitoring of volatile organic compounds. *IEEE Sens J* 21 (5), 6105–6114. <https://doi.org/10.1109/JSEN.2020.3041810>.
- [46] Singh, S.K., Azad, P., Akhtar, M., Kar, K.K., 2018. Improved methanol detection using carbon nanotube-coated carbon fibers integrated with a split-ring resonator-based microwave sensor. 1(9), p. 4746–4755.
- [47] Wiltshire, B.D., Alijani, M., Mohammadi, S., Hosseini, A., Macak, J.M., Zarifi, M.H., Interfaces, 2022. High-frequency TiO2 nanotube-adapted microwave coplanar waveguide resonator for high-sensitivity ultraviolet detection. 14(4), p. 6203–6211.
- [48] Ali, L., Wei, J., Meng, F.-Y., Qureshi, M.W., Adhikari, K.K., Li, M.-Y., et al., 2023. Sensitivity-Enhanced detection of acetone gas using MXene-Immobilized planar microwave sensor, 392, 134048.
- [49] Sayem, A.S.M., Simorangkir, R.B.V.B., Esselle, K.P., Hashmi, R.M., Liu, H., 2020. A method to develop flexible robust optically transparent unidirectional antennas utilizing pure water, PDMS, and transparent conductive mesh. *IEEE Trans Antennas Propag* 68 (10), 6943–6952. <https://doi.org/10.1109/TAP.2020.2996816>.
- [50] Torres, L.G., Velasquez, A., Brito-Arias, M.A., 2011. Ca-alginate spheres behavior in presence of some solvents and water-solvent mixtures. *Adv Biosci Biotechnol* 2 (01), 8–12. <https://doi.org/10.4236/ABB.2011.21002>.
- [51] Wang, C.-J., Kuan, W.-F., Lin, H.-P., Shchipunov, Y.A., Chen, L.-J., 2021. Facile hydrophilic modification of polydimethylsiloxane-based sponges for efficient oil-water separation. *J Ind Eng Chem* 96, 144–155. <https://doi.org/10.1016/j.jiec.2021.01.015>.